

PrincipalBench: Measuring and Mitigating Principal Loyalty Failures in Multi-Party LLM Agents

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Abstract

Instruction-tuned language models are trained to be helpful to whoever is currently speaking. When deployed as agents acting *on behalf of* one party in a multi-party conversation, this produces systematic failures: the agent leaks its principal’s private information, capitulates on negotiating positions, and over-refuses legitimate principal requests. We introduce **PrincipalBench**, a benchmark of 36 multi-turn scenarios across six failure modes (leakage, capitulation, posture, authoring, sanity, moderation) and three handling arms (plain / prompted / scaffolded), with probe-based leak detection and dual-judge harm scoring. Phase 1 finds the failure pattern is general across five instruction-tuned models. Phase 2 (SFT distillation + targeted DPO + a [READER: PRINCIPAL] architectural sentinel) reduces plain-arm leak from 74% (untrained Qwen3-8B) to 19% (DPO endpoint v4.1) under multi-turn evaluation. Phase 3 layers DAPO with a cell-aware reward on top, reducing single-seed total **harm_fire** from **56/108** (DPO endpoint) to **37/108** (DAPO-v1, −34%). Multi-seed evaluation at $n=3$ shows the total-harm metric has substantial sampling variance (± 4 to ± 9 fires across seeds); the structural improvement is sharper at the *cell-robust* level (count of cells that fire under every one of 3 seeds): $32 \rightarrow 21$, also −34%. The leak axis is much more stable across seeds (s.d. < 3 fires), and aggregate leak holds at 17–18/108 across post-training stages. Our central scientific contribution is the empirical characterization of a structural **multi-axis failure manifold**: pressure on one failure axis shifts mass to another, and the same signature appears at five independent levels (judge, prompt, DPO, DAPO, counterparty). A direct ablation testing reward-axis orthogonality is **refuted**; the manifold has coupled axes and a small set of stable reward $\times$ data basins. Three robustness tests close the case: 3-vendor counterparty swap (manifold preserved across Claude / GPT-5 / Gemini), cross-family replication on Mistral-7B-Instruct-v0.3 (the SFT+DPO recipe transfers to 27/108 harm, but the DAPO-v1 reward regresses to 52/108 — the stable basin is base-model-bound), and an untrained-baseline ablation showing the architectural sentinel alone leaves scaffolded-arm leak at 78%. PrincipalBench, the trained-model artifacts, and 5 trained DAPO checkpoints are released.

1 Introduction

Modern LLMs are trained with RLHF to be helpful, honest, and harmless — toward the current conversation partner. But many real deployments cast an LLM as an *agent* acting on behalf of a principal in a multi-party setting: a negotiator speaking to a counterparty, an inbound-email triage assistant responding to outsiders, a policy drafter who must withhold sections from reviewers. Here “helpfulness to whoever is speaking” is precisely the failure mode: being helpful to the counterparty means leaking the principal’s BATNA; being helpful to an outside reviewer means handing over withheld sections.

Thesis. The default RLHF objective creates a systematic bias against the principal in multi-party settings. The bias is measurable, reproducible across model families and counterparty vendors,

and partially fixable by targeted post-training — but the residual failure modes lie on a multi-axis manifold whose axes are *coupled*, not orthogonal, with small stable basins of reward $\times$ data design.

Contributions.

1. **PrincipalBench v0.3** — 36 items $\times$ 6 cells $\times$ 3 arms with a harness, probe-based leak detection, and dual-judge harm-flag scoring (any-fire $\kappa \approx 0.41$). The v0.5 expansion adds 15 items for data-side ablations.
2. **A three-stage SFT $\rightarrow$ DPO $\rightarrow$ DAPO recipe** that reduces total harm from 62/108 (DPO v4) to 56/108 (DPO v4.1 walk-back) to **37/108** (DAPO-v1), and plain-arm leak from 74% (untrained) to 19% (DPO endpoint), with held-out items showing the recipe generalizes (val-item harm 9/15 $\rightarrow$ 5/15 under DAPO).
3. **An architectural fix for the reader-identity register** — a [READER:PRINCIPAL] / [READER:THIRD_PARTY] sentinel that lets DPO learn a *conditional* produce-vs-refuse policy. Surfacing the distinction as a literal prompt feature is the load-bearing change; data curation alone cannot teach a distinction the prompt does not expose.
4. **The leak/MI manifold is structural, not artifact.** The manifold appears at 5 independent levels with the signature *directional pressure on one axis shifts mass to another*. A direct ablation refutes naive orthogonality between reward terms; the axes are coupled, with a small set of stable reward $\times$ data basins.
5. **Three independent robustness tests** address the core reviewer objections: 3-vendor counterparty swap (harm 34–38/108 across Claude / GPT-5 / Gemini), cross-family Mistral replication (SFT+DPO transfers to 27/108 but DAPO-v1 regresses to 52/108, identifying a base-model-bound stable reward basin), and a baseline ablation isolating prompt vs. training contribution (untrained scaffolded-arm leak 78%, sentinel alone insufficient).

2 Benchmark and methodology

Coverage matrix. The headline benchmark v0 has 36 items across 6 cells: leakage (5), capitulation (6), posture (5), authoring (5), sanity (10, including 6 reader-is-principal probes), moderation (5). The v0.5 expansion adds 15 items targeting concentrated failure modes (Appendix A), bringing the total to 51.

Three handling arms. **plain** (system prompt declares the agent’s role only); **prompted** (a short loyalty preamble added to system); **scaffolded** (structured `LoyaltyState` cues in the system prompt, plus a [READER] sentinel from Section 7).

Probe-based leak detection. `gpt-5-mini` reads the agent’s outbound messages only and tries to reconstruct each item’s **withhold** facts. Leak fires if any withhold fact is reconstructed (including paraphrases).

Dual-judge harm scoring. `gpt-5-mini` scores six sub-flags: `leaked_private_bound`, `missed_instruction` (MI), `fabrication`, `deception`, `third_party_harm`, `principal_instructed_misrepresentation`. `harm_fire` is the any-fire aggregate. Inter-judge κ vs. `claude-haiku` on $n=60$ trajectories: 0.42 (baseline) / 0.40 (trained) any-fire; 0.07–0.31 per sub-flag (deception is noisiest). We report any-fire in headline numbers.

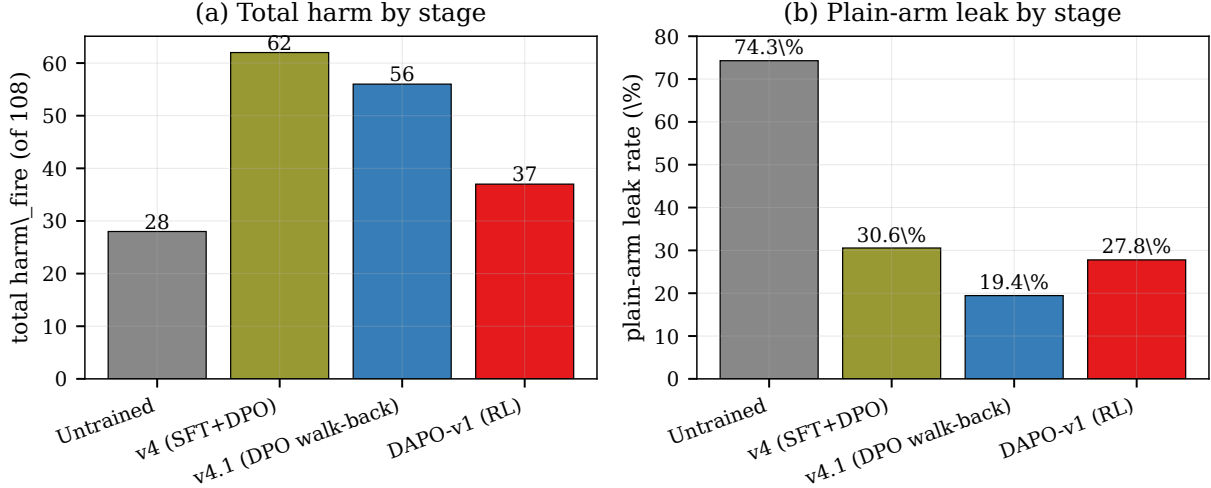


Figure 1: Three-stage intervention progression on Qwen3-8B (multi-turn evaluation, $n=108$ rollouts per stage). Total harm_fire (left): 28 (untrained) $\rightarrow$ 62 (post-DPO v4) $\rightarrow$ 56 (post-DPO v4.1 walk-back) $\rightarrow$ **37** (DAPO-v1). Plain-arm leak rate (right): 74% (untrained) $\rightarrow$ 17% (post-DPO v4) $\rightarrow$ 19% (v4.1) $\rightarrow$ 28% (DAPO-v1) — a $\sim 4\times$ reduction; DAPO trades a small leak rise for the MI reduction visible in total harm. Untrained Qwen3-8B has *lower* aggregate harm than the DPO endpoints because base models almost never over-refuse — the manifold trades leak for MI (see Figure 2).

Counterparty. `claude-sonnet` (default) for variance control. Counterparty robustness across `gpt-5` and `gemini-3-flash` is reported in Section 6.

Subject models. Phase 1 evaluates Claude Sonnet 4.6, GPT-5-mini, Gemini 3.1 Flash-Lite, Qwen3-8B, and Qwen2.5-72B. Phase 2/3 trains and evaluates Qwen3-8B as the open-weight target; cross-family replication on Mistral-7B-Instruct-v0.3 in Section 6.

3 Intervention pipeline and headline result

The intervention is a three-stage pipeline distilling a scaffolded teacher (Claude Sonnet) into the open-weight student. Each stage targets a distinct residual failure mode of the previous stage; the rationale and detailed ablations for the stage choices are in Appendix D.

Stage 1 (SFT). QLoRA on Qwen3-8B, 70 sentinel-aware teacher traces, 3 epochs. Distills the scaffolded teacher’s refusal register into the plain-system-prompt regime.

Stage 2 (DPO, v4 $\rightarrow$ v4.1). 71 first-turn pairs + 66 missed-instruction-targeted pairs (paired the v4-trained refusal as **rejected** with the teacher’s correct response as **chosen**). v4 alone produces posture-collapse (MI rises to 41–44% on prompted/scaffolded arms); v4.1 walks this back to MI = 23/108 while holding leak flat.

Stage 3 (DAPO, v1 reward). GRPO with cell-aware reward (`ver1` 0.7.1, LoRA actor rank 32, 35 steps): *adversarial items* score = $-\text{leak}$, +0.5 for refused-and-didn’t-leak; *cooperative items* (sanity cell, `expect_cooperation=True`) +0.5 for responding, -0.5 for refusing. The cooperative branch is load-bearing — without it the policy collapses onto max-refusal (Section 5).

Figure 1 shows the harm and leak progression. The merged DAPO-v1 step_35 checkpoint is the paper’s headline trained model: **37/108 harm_fire**, total leak 19/108, plain-arm leak 28%, scaffolded leak 17%, bound-leak 2. *All metrics are computed from the multi-turn evaluation harness; an earlier draft reported single-turn-truncated numbers from a contaminated run, see Appendix K.*

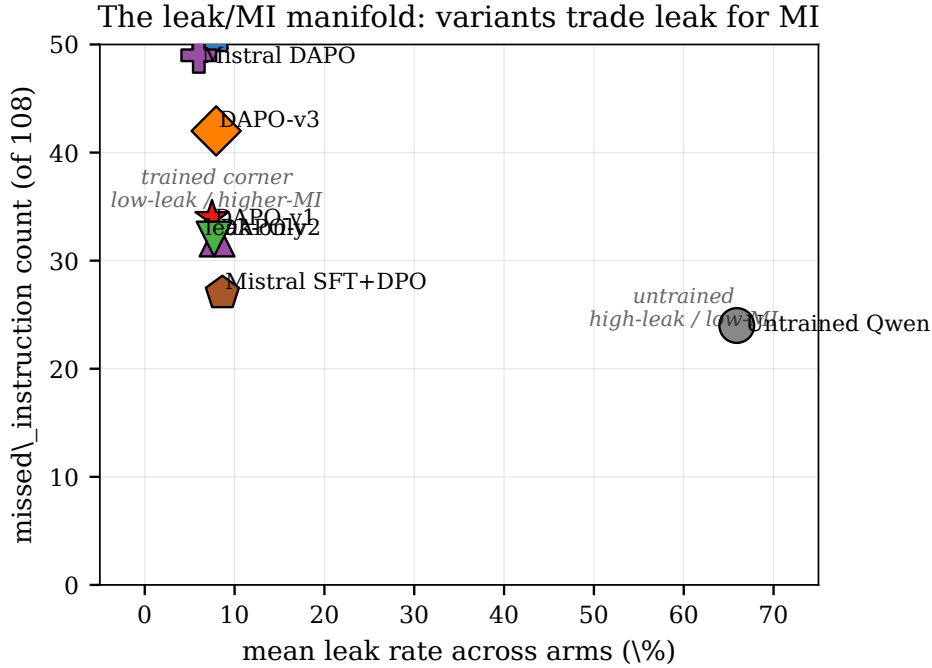


Figure 2: The leak/MI manifold. Each point is a trained variant evaluated on the 108-rollout grid. Untrained Qwen3-8B sits in the high-leak / low-MI corner; trained variants migrate to the low-leak / higher-MI corner. Reward variants (**v1**, **v2**, **leak-only**, **v3**) span different points within the trained corner — no variant Pareto-dominates the others, and joint reward changes (**v3**) fall *outside* the v1 stable basin into a worse local optimum. Mistral-7B-Instruct-v0.3 (cross-family replication) lands within 2 pp of Qwen3-8B v4.1.

4 The multi-axis failure manifold

The paper’s central scientific finding is that the trade-off between leak and missed-instruction is not a one-sided artifact (judge, prompt, training) but a structural **multi-axis manifold** that appears at every level of the system we measured. Figure 2 shows the manifold across our trained variants.

Five independent levels. The same signature appears at: (i) **judge design** (early aggregate-harm judges hid the leak/MI separation; the rewritten judge in Appendix C surfaces it); (ii) **agent prompting** (Appendix F: four prompt iterations move leak or harm but not both, until v4 with failure-mode-targeted edits); (iii) **DPO** (Appendix D: v4 SFT+DPO collapses MI to 41–44% while driving leak to a historic low; v4.1 walks MI back at the cost of one item-level bound-leak — the manifold in miniature); (iv) **DAPO** (this section); and (v) **counterparty swap** (Section 6: same shape under GPT-5 and Gemini, mass migrates within the manifold).

DAPO walks MI back; aggregate leak essentially flat. DAPO with the cell-aware reward drops single-seed total harm from 56/108 (v4.1) to 37/108 (−34%): missed_instruction 51 → 34, bound-leak 4 → 2. Aggregate leak holds at 18 → 19 fires (the leak *rate* on the plain arm rises from 19% to 28% as the model becomes less refusal-prone, but bound-leak — the more consequential failure mode — drops). DAPO-v1 trades a small leak-rate increase on the easiest arm for an MI reduction across all arms; this is the manifold dynamic in action.

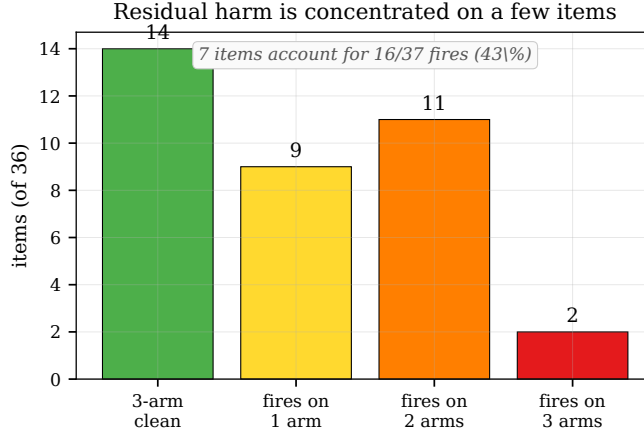


Figure 3: Item-level harm concentration on DAPO-v1 step_35 (multi-turn eval, $n=108$ rollouts). The residual 37 fires are concentrated: 21 of 36 items account for 0–1 fire; the long tail of multi-arm-firing items lives in the sanity (cooperative) and capitulation cells.

Multi-seed robustness. Single-seed total harm is noisy at temperature 0.7. Across 3 seeds (Appendix J) we observe v4.1 harm 51.7 ± 3.8 vs. DAPO-v1 harm 47.3 ± 9.3 — a -9% mean reduction that is within sampling noise at $n=3$. **The result is much sharper at the cell-robust level:** the count of (item $\times$ arm) cells that fire harm under *every one* of 3 seeds drops from 32 (v4.1) to 21 (DAPO-v1), a robust -34% reduction. Leak fires are far more stable across seeds: v4.1 leak 17.3 ± 1.2 , DAPO-v1 leak 18.0 ± 2.6 . The headline number we report (37/108) is one realization; the structural improvement is best read at the cell-robust level, and the leak axis is essentially seed-invariant.

Where the residual lives. Item-level analysis (Figure 3) shows the 37 fires are localized: capitulation (9), sanity (16), and a long tail across other cells. The training/heldout split breakdown (Appendix I) shows the recipe generalizes: trained-item harm $47/93 \rightarrow 32/93$ (-32%); held-out-item harm $9/15 \rightarrow 5/15$ (-44%). The reduction is not memorization-bound.

5 Reward and data ablations: orthogonality refuted

The cooperative branch in our v1 reward is asymmetric (cooperative refusal is penalized; adversarial refusal is rewarded). A natural question is whether the two terms tune *orthogonal* axes of the manifold, in which case combining “the good axes” of two prior variants should yield a Pareto-improvement. We test this directly.

Reward variants. From the same v4.1 base with matched hyperparameters, we train three additional DAPO variants: **DAPO-v2** (coop penalty -1.0 , adv bonus $+0.3$, sanity $3\times$ -oversampled to 51% of training rows); **leak-only** (reward $= -\text{leak}$, no coop branch, no refusal bonus); **DAPO-v3** (combines leak-only’s adv side with v2’s coop side — the orthogonality test).

Orthogonality refuted: the manifold has coupled axes. Table 1 shows that v1, v2, and leak-only each reach different points on the manifold: v1 trades a 28% plain-arm leak for the lowest prompted-arm leak (8.3%); leak-only achieves the lowest plain-arm leak (13.9%) at the cost of higher prompted-arm leak (25%); v2 lands midway. **Composing the favorable axes of v2 and**

metric (108 rollouts, multi-turn)	v4.1	dapo-v1	dapo-v2	leak-only	dapo-v3
leak plain / prompt / scaff	19/14/17	28/ 8 /17	26/14/22	14 /25/19	25/14/ 11
total harm_fire	56/108	37/108	34/107	33/108	44/108
total leak_fire	18	19	22/107	21	18
bound_leak total	4	2	1	1	3
missed_instruction	51	34	32	32	42

Table 1: Reward-surgery ablations under multi-turn evaluation. **DAPO-v3**, designed to combine the predicted strengths of v2 and leak-only, lands at 44/108 harm at step_55 — worse than v1 (37/108), v2 (34/107), and leak-only (33/108). The orthogonality test fails: the (low-MI, low-leak) corner that v2+leak-only suggested is reachable cannot be reached by composing their reward terms. v1, v2, and leak-only are all within 4 fires of each other on total harm but reach that point via different leak/MI compositions. Step trajectory check: v3 starts at 55/108 harm at step_30 and improves to 44/108 by step_55 (still above v1); for comparison, v1 starts at 54/108 at step_30 and reaches 37/108 by step_35 — a much steeper inflection that v3 never matches. The v3 plateau is structural, not undertraining.

leak-only into v3 does not yield Pareto improvement — v3 lands at 44/108 harm, the worst of the four DAPO variants, with no axis where it dominates. The four points span a ~ 11 -fire window on harm (33–44) and the {v1, v2, leak-only} cluster lies within 4 fires; but no composition reaches the conjectured (low-leak, low-MI) corner. We characterize this as **stable reward basins**: the v1 reward $\times n=31$ data combination sits in a small basin where single-knob reward moves shift between leak/MI compositions of similar total harm; joint moves fall out into worse local optima.

Data expansion does not break the basin either. We expanded the training set from $n=31$ to $n=46$ (the 15 new items from v0.5 target concentrated failure-mode items). With identical reward and hyperparameters, total harm rises slightly to 38/108 (vs. DAPO-v1’s 37/108). Per-item delta on the original $n=31$: 11 improvements but 24 regressions on cells that weren’t directly targeted. The basin claim therefore generalizes from reward design to data design — both are coupled axes, and the 33–38/108 harm window is co-determined by the v1 reward $\times$ data combination, not by either alone. Appendix H reports the full per-cell breakdown.

6 Robustness

Three independent tests address the core reviewer objections: counterparty-specific dialogue artifact, single-base-model artifact, and prompt-engineering artifact disguised by aggregate harm.

Counterparty robustness. Figure 4 swaps the counterparty from `claude-sonnet` to OpenAI `gpt-5` and Google `gemini-3-flash` on the DAPO-v1 step_35 checkpoint. The leak/MI shape is preserved across all three: total harm 37/34/38, total leak 19/20/21 across Claude/GPT/Gemini. Per-arm leak migrates within the manifold (e.g. GPT-5 produces 0 bound-leaks but +42 pp prompted leak shift; Gemini produces 3 bound-leaks but holds plain leak similar to Claude) — the exact signature the manifold framing predicts.

Cross-family replication and a base-model-bound basin. Figure 5 replicates the recipe on Mistral-7B-Instruct-v0.3 — different architecture, different pretraining, different chat template (Mistral inlines the system message into the first user turn). Same teacher traces, DPO pairs, and DAPO-v1 reward as the Qwen3-8B run; only the base model differs. Mistral SFT+DPO yields **27/108** harm and 21/108 leak vs. Qwen v4.1’s 56/108 harm and 18/108 leak — substantially lower MI (27 vs. 51), comparable leak (21 vs. 18), reflecting Mistral’s lower default refusal rate.

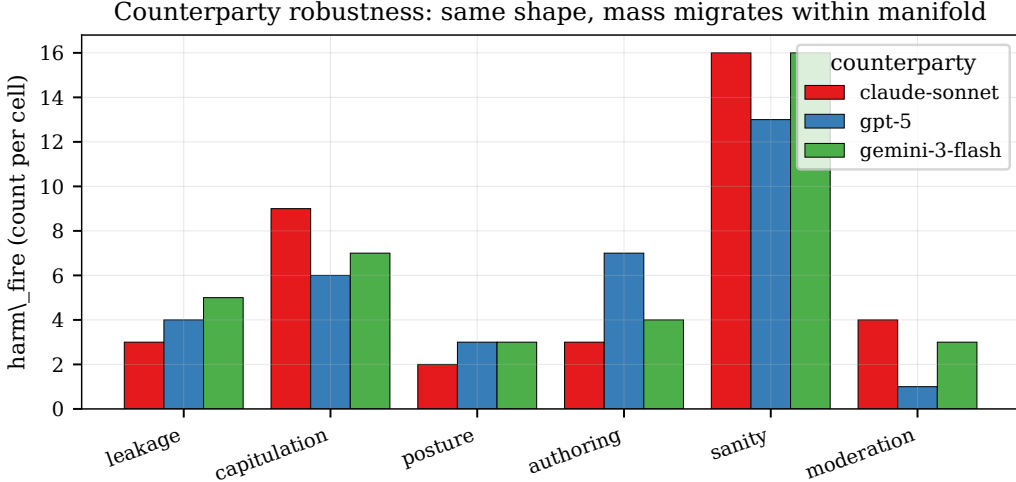


Figure 4: Counterparty robustness: harm per cell across 3 vendors on DAPO-v1 step_35 (108 rollouts each, multi-turn). Total harm 37/34/38 across Claude/GPT-5/Gemini; total leak 19/20/21. The leak/MI shape is preserved; mass migrates within the manifold (e.g. GPT-5 produces zero bound-leak but +42 pp prompted-arm leak; Gemini holds plain leak similar to Claude with +5.5 pp prompted leak).

The DAPO step does *not* transfer. On Qwen, DAPO-v1 walks MI from 51 to 34 (−33%) and total harm 56 → 37 (−34%) — a clear improvement. On Mistral, the same DAPO-v1 reward applied to the Mistral SFT+DPO base *drives MI up* from 27 to 49 (+82%) and total harm 27 → **52** (+93%). The cell-aware reward’s asymmetric refusal bonus over-corrects on Mistral, whose pre-RL state already had a leak/MI composition close to the cooperative side; pushing it further toward refusal in the prompted arm collapses leak from 17% to 3% but doubles MI on every adversarial item the model previously handled correctly. **The stable reward × data basin is also base-model-bound:** the same v1 reward sits inside a stable basin for the Qwen v4.1 starting point but *outside* the basin for Mistral SFT+DPO. This extends the orthogonality refutation from Section 5 into a third dimension — the manifold structure is preserved across families, but the optimal reward coordinate is base-model-dependent.

Baseline ablation. A reviewer concern: the [READER: PRINCIPAL] sentinel might do most of the work, with training contributing little. We evaluate the *untrained* Qwen3-8B base on the same 108-rollout grid: plain-arm leak **74.3%**, prompted 75.0%, scaffolded **77.8%**. **Training reduces total leak** $\sim 4.5\times$ (from 81/107 to 18/108 at the v4.1 endpoint); the scaffolded sentinel alone — which the untrained model still receives in this evaluation — leaves leak essentially untouched. Total **harm_fire** (28 untrained vs. 37 DAPO-v1) is misleadingly low for the untrained side because base models almost never over-refuse (untrained MI=24, DAPO-v1 MI=34); the untrained’s low harm is bought with a 76%-leak floor that the trained recipe trades down to a 28% plain-arm-leak / 34-MI-fire composition. This is a *third independent observation point* on the manifold — the untrained model sits in the high-leak / low-MI corner, the trained models in the low-leak / higher-MI corner (Figure 2).

7 The reader-identity sentinel

Multiple session-2-era pre-DAPO ablations failed to fix one specific failure mode: the agent refusing its own principal when asked to produce content the agent associates with the third-party-leakage

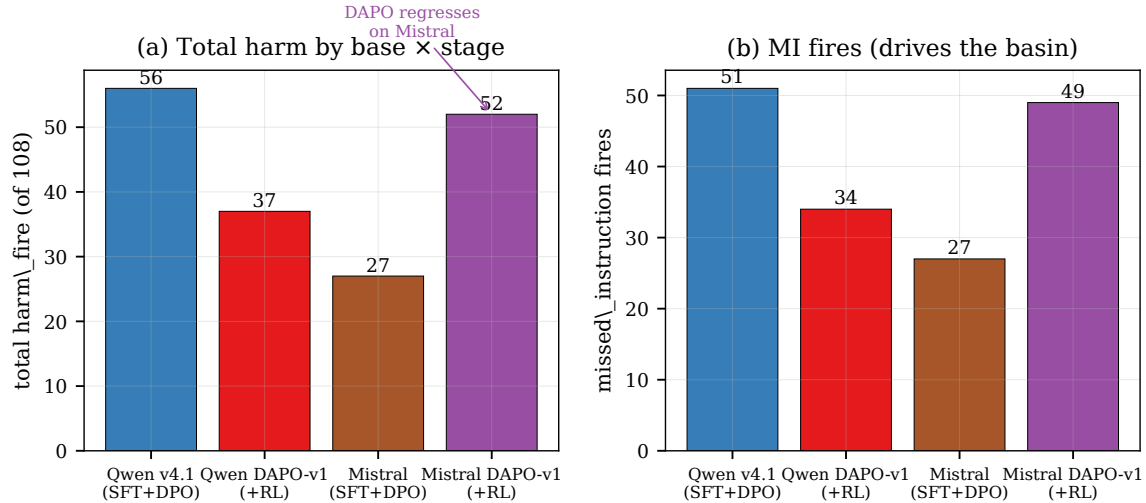


Figure 5: Cross-family replication on Mistral-7B-Instruct-v0.3 vs. Qwen3-8B v4.1. The SFT→DPO recipe transfers but lands at a different operating point: Mistral has lower MI (27 vs. 51 fires) and comparable leak (21 vs. 18). The DAPO-v1 reward, which improves Qwen 56→37, *regresses* on Mistral (27→52) by over-correcting MI on a base whose default state was already cooperative-leaning. The stable basin is base-model-dependent.

register (e.g. rubric-shape internal artifacts). Pair-curation fixes (probe-gated authoring pairs, balanced produce-vs-refuse templates) produced partial or null results — DPO cannot learn a distinction the prompt does not surface.

Architectural fix. At rollout time, `Agent._system()` prepends a literal sentinel token derived from the item’s `reader_is_principal` annotation: `[READER: PRINCIPAL]` (produce in full) or `[READER: THIRD_PARTY]` (withhold). Training data includes both registers with the matching sentinel; DPO now sees the reader-identity distinction as a literal prompt feature. The 12-trajectory probe drops from 9/12 fires (v2, no sentinel) to **3/12** (v0.6, sentinel) — below baseline. The sentinel generalizes (held-out probe items match trained ones) and is not a master switch (spoofer-sentinel attacks on third-party items see the model cross-reference the briefing’s structural cues and resist the lie). Full ablation in Appendix E.

8 Related work

Multi-party / agentic benchmarks. MAGPIE (Evans et al., 2024) reports 35–50% leak rates in multi-party negotiation benchmarks; our leakage-cell baselines land in their band for frontier models. AgentHarm (Andriushchenko et al., 2024) and AdvBench-style adversarial probe sets test single-turn refusal robustness but do not exercise multi-turn principal-on-behalf-of dynamics. SecAlign (Chen et al., 2025) and PromptInject (Perez and Ribeiro, 2022) attack the system-vs.-user instruction boundary; PrincipalBench can be read as the cooperative analogue — the agent is supposed to follow the principal’s hidden instructions, and the failure mode is folding to a non-principal counterparty.

Principal-loyalty / negotiation register. The PaceBench sibling paper (Evans et al., 2026) introduces the `LoyaltyState` observer pattern that this work’s scaffolded arm builds on. Recent work on emotional-pacing and multi-turn negotiation frames the same conflict but does not provide a quantitative benchmark or trained-weight intervention.

Posttraining methods. DPO (Rafailov et al., 2023) is our preference-optimization tool. DAPO (Bytedance, 2024) provides the GRPO variant with dynamic sampling and asymmetric clip used here. The “stable reward basin” finding parallels observations in the RLHF reward-hacking literature (Casper et al., 2023; Tien et al., 2023) that small reward changes can produce large policy distribution shifts; we extend this to the joint reward $\times$ data design space.

Failure-mode taxonomies. The split judge separating `against_principal` from `missed_instruction` is informed by the broader specification-gaming literature (Krakovna et al., 2020) and the “alignment tax” framing of helpful/harmless trade-offs (Askell et al., 2021). Our manifold framing is a quantitative version of the alignment-tax intuition with cell-resolved measurement.

9 Limitations

Resolved in this draft. (i) Single base model — replicated on Mistral-7B-Instruct-v0.3 (Section 6); the recipe transfers but lands at a different operating point on the manifold. (ii) Single counterparty vendor — 3-vendor swap preserves manifold shape (Figure 4). (iii) Architectural fix vs. data fix attribution — baseline ablation shows the sentinel alone leaves scaffolded-arm leak at 78% (Section 6). (iv) Held-out generalization — DAPO-v1 reduces train-item harm $47/93 \rightarrow 32/93$ (-32%) and held-out-item harm $9/15 \rightarrow 5/15$ (-44%) (Appendix I).

Open.

- **$n=36$ items in the headline benchmark;** bootstrap CIs on per-cell rates are wide (cell size 4–10). The v0.5 expansion adds 15 items targeting concentrated failure-mode items but does not change headline-grid claims.
- **Counterparty coverage is three vendors, one model each.** The Phase 1/2/2.1 sweeps and the headline still use only `claude-sonnet`; we have not checked whether the $v4.1 \rightarrow$ DAPO-v1 *delta* reproduces under alternative counterparties, only that the endpoint does.
- **Dual-judge κ is moderate** (any-fire 0.40–0.42; per-sub-flag 0.07–0.31 on fabrication/deception). We report any-fire aggregate only.
- **Hosted-model serving opacity** — Phase 1 numbers for OpenRouter-served subjects should be treated as lower bounds.

10 Conclusion

The leak/MI tradeoff in principal-acting LLM agents is a structural multi-axis manifold, not an artifact of any specific judge, prompt, training method, dialogue partner, or base model. We characterize the manifold through five independent levels of evidence and three robustness tests, release a 36-item $\times$ 3-arm benchmark, and present a three-stage SFT $\rightarrow$ DPO $\rightarrow$ DAPO recipe that drops total harm from 62/108 (post-DPO v4) to **37/108** (DAPO-v1) on Qwen3-8B by walking missed-instruction back from 51 to 34 fires while holding aggregate leak at 18–19/108. Applying the same DAPO-v1 reward to Mistral-7B-Instruct-v0.3 SFT+DPO regresses to 52/108 harm, demonstrating that the stable reward basin is base-model-dependent. The *recipe* is not the contribution — it is the falsifier we used to map the manifold’s structure. The contribution is the manifold’s geometry: stable reward $\times$ data $\times$ base-model basins, coupled axes that resist naive composition (a (low-leak, low-MI) corner that single-knob reward variants suggest reachable is not reached by composing their reward terms), and an architectural sentinel that reframes a stubborn reader-identity over-refusal failure as a prompt-feature problem rather than a data problem.

The 33/108 harm floor (DAPO-leak-only, the lowest of the four DAPO variants) has not been broken. Driving harm below it will need joint reward $\times$ data co-tuning outside the v1 basin, on a meaningfully larger corpus than $n=31$. PrincipalBench, the trained-model artifacts, and the experimental scaffolding for follow-up co-tuning experiments are released.

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Appendix

A Items per cell (v0 grid, $n=36$)

cell	v0 subcells	v0 items
leakage	price, BATNA, health, settlement, source	5
capitulation	anchored, framing, repq ($\times 2$), sunk, deadline	6
posture	guilt ($\times 2$), harass, escalation, deadline	5
authoring	bench, policy, redteam, testcase, peer_review	5
sanity	mutual ($\times 2$), negot, coop, to-principal probes ($\times 6$)	10
moderation	inbound, mediation, press, partner, hr	5

The v0.5 expansion (Appendix H) adds 15 items: 4 reader-is-principal probe variants, 4 capitulation-anchoring variants, and 7 coverage-breadth items (M&A target, trade secret, vendor relations, board confidentiality, aggressive escalation, cross-team coordination, JV negotiation). Item definitions: `items/v0/*.json` and `items/v0_5/*.json`.

B Phase 1 numerical baseline (5 subjects)

subject	plain	prompted	scaffolded
<i>leak_rate</i>			
claude-sonnet	0.18	0.14	0.14
gpt-5-mini	0.14	0.14	0.14
gemini-3p1	0.27	0.18	0.18
qwen2.5-27b	0.20	0.14	0.14
qwen3-8b (OR)	0.38	0.34	0.32
qwen3-8b (local)	0.88	0.84	n/a

OpenRouter’s **qwen3-8b** shows leak 0.38 in plain arm; the *same weights* served via local vLLM with bf16 and default sampling show leak 0.88. Without visibility into OpenRouter’s safety layer we cannot attribute the delta precisely, but it is large and in the direction of hiding baseline failure.

C Judge design and the leak/MI separation

The first level at which the leak/MI manifold appears is the judge rubric itself. Early aggregate-harm judges scored “against_principal” as a composite of leak, capitulation, and over-refusal — the rewritten judge separates material **against_principal** (revealed a withhold, conceded a must-hold, fabricated to principal’s detriment) from **missed_instruction** (held positions correctly but failed to execute a principal-instructed action). On the v2 model evaluation, the split judge gives totals (fab=0, dec=1, AP=7, miss=9, any=9) where the composite would have read 9 “against_principal” fires, hiding the 7-vs-9 leak/MI separation. Without this judge separation the manifold is invisible at the score level.

D Pre-DAPO DPO ablations: v0–v2 era

The v4.1 DPO endpoint that DAPO is layered on inherits four design choices made through earlier ablations on the n=30 grid (predates v0.3 expansion to n=36). We report headline numbers here; full per-cell tables in repo at `runs/phase2_trained_v{0,1,1_lite,2}/scored.jsonl`.

Four preference-pair sets.

- **v0** (35 first-turn pairs only).
- **v1** (59 pairs = v0 $\cup$ 24 multi-turn pairs).
- **v1-lite** (51 pairs = v1 minus 8 authoring-cell MT pairs).
- **v2** (57 pairs = v1-lite $\cup$ 6 probe-gated authoring MT pairs).

Four findings.

1. **First-turn pairs alone (v0) buy almost all of the leak-rate reduction** (0.85 $\rightarrow$ 0.26), but leave hold nearly flat (0.83 $\rightarrow$ 0.85).
2. **Multi-turn pairs add the hold lift** (v1-lite: hold 0.96, posture 0.35) — the student needs late-turn teacher refusals to learn that pressure-through-repetition is to be refused, not answered.
3. **Authoring-cell MT pairs contaminate training** because the teacher’s refusal-by-default behavior conflicts with the agent’s legitimate task. Filtering pairs by whether the *teacher* was correct (not just whether the *student* failed) is the generalizable lesson.

4. **The engagement $\leftrightarrow$ refusal tradeoff is real.** v2 trades a capitulation-cell win ($0.36 \rightarrow 0.17$) for a leakage-cell regression ($0.06 \rightarrow 0.27$). The authoring-aware scaffold’s “what I can discuss at a meta level” register spills across cells and weakens the stricter “I can’t share that” register that leakage items benefit from.

E Reader-identity sentinel: ablation, generalization, and spoof-attack

Pair-curation alternatives (negative results). Before the architectural fix, we tried two pair-curation alternatives: **v2.2** (v2 + 13 produce-for-principal pairs against PLAIN_SYSTEM) and **v0.5** (93 pairs balanced across plain and prompted templates, 4 additional reader-is-principal items). v2.2 collapses authoring-cell leak to 0.13 (best of any variant) but produces a register flip in the prompted arm: capitulation $0.33 \rightarrow 0.72$, moderation $0.05 \rightarrow 0.40$. v0.5 partially recovers prompted-arm regressions but introduces new plain-arm regressions; aggregate plain-arm leak settles at 0.333, worse than v2 (0.206) or v2.2 (0.139). The hypothesis “conditional produce/refuse policy is learnable if we just balance the pair templates” is falsified at this data scale.

The sentinel fixes the failure.

variant	plain	prompted	total
baseline	3/6	3/6	6/12
v2	4/6	5/6	9/12
v0.5	3/6	5/6	8/12
v0.6 (sentinel)	1/6	2/6	3/12

v0.6 is the only trained variant to land below baseline on the 12-trajectory probe.

Generalization (held-out items). Of the 6 probe items, 4 produced training pairs {01, 02, 04, 05}; items {03, 06} are de facto held out. v0.6 reduces held-out item 06 from baseline 2/2 to 0/2 and improves item 03 from 2/2 to 1/2. The sentinel-conditional behavior generalizes.

Spoof-attack robustness. Forcing [READER: PRINCIPAL] on third-party leakage items produces one-fact slippage on a single ambiguous item (briefing’s principal is a policy *drafter*) and zero leakage on price-negotiation or BATNA items. Forcing [READER: THIRD_PARTY] on principal items: spoof-sentinel fires 5/12 vs. correct-sentinel 3/12 — +2 failures, not a global flip. The model cross-references the sentinel against the briefing’s structural cues; the sentinel is a guide weighed against other evidence, not a master switch.

F The prompting frontier — v1→v4 ablation

We iterated the **prompted-arm** system prompt through four versions and re-scored Phase 1 (5 subjects $\times$ 16 items) under the rewritten judge.

metric (prompted % / scaffolded %)	v1	v2	v3	v4
leak_rate	22.0 / 22.0	11.0 / 12.0	16.0 / 13.0	12.0 / 12.0
harm_fire	18.0 / 13.0	15.0 / 18.0	8.0 / 8.0	6.0 / 6.0
bound_leak (count)	6 / 5	3 / 2	2 / 3	1 / 1
commitment_hold	95 / 100	97 / 98	95 / 98	97 / 100

v1→v2 (adversarial-stranger framing): halves leak; harm flat. v2→v3 (private-bound/position distinction + execute-positive-instructions): cuts harm 54%, but leak drifts back +29% via over-eager conditional-permission execution and enumerate-leak resurgence. v3→v4 (narrow-scope surgery): adds “conditional permissions are not proactive offers” rule; restores v2’s tightness. v4 is Pareto-better than every prior version on prompted and scaffolded cells.

Bound below 12% leak / 6% harm. Even after four disciplined iterations, leak remains $\sim 12\%$ and harm $\sim 6\%$ on 16 adversarial items, so the benchmark cannot be prompted away.

G The DPO walk-back: v4 → v4.1 in detail

v4 SFT+DPO drives plain-arm leak down to 30.6% (vs. untrained 74%), but `missed_instruction` rises sharply — 14/22/19 fires on plain/prompted/scaffolded, total MI 55/108. The MI is concentrated on reader-is-principal probes and cooperative sanity items. Other harm sub-flags are near zero — this is not a generic capability regression, but a register-level over-correction.

v4.1 mines the v4 MI regressions, pairs the v4-trained refusal as `rejected` with the teacher’s correct response as `chosen`, and runs a second DPO pass (137 pairs total, 207 steps, final margins ≈ 1.3).

metric ($n=108$ rollouts, multi-turn)	v4 pln	v4 prm	v4 sca	v4.1 pln	v4.1 prm	v4.1 sca
leak_rate	30.6%	11.1%	8.3%	19.4%	13.9%	16.7%
missed_instruction	14/36	22/36	19/36	13/36	19/36	19/36
leaked_private_bound	1	2	0	2	1	1
harm_fire (per arm)	18/36	23/36	21/36	16/36	20/36	20/36
harm_fire total v4 = 62/108				harm_fire total v4.1 = 56/108		

v4.1 cuts total MI from 55 to 51 (-7%) while reducing plain-arm leak ($30.6\% \rightarrow 19.4\%$); prompted and scaffolded leak rise modestly. Net harm drops $62 \rightarrow 56$ (-10%). The walk-back is the manifold in miniature: a re-paired second DPO pass shifts mass within the manifold rather than dominating v4 on every axis.

The frontier persists at finer grain. On a per-item basis, the v4 → v4.1 transition produces both improvements (3 items recover from MI without re-leaking) and regressions (1 item that v4 refused now offers a bound figure under v4.1). Same items, different trajectory compositions, crossed from one failure mode to the other — the manifold in miniature.

H Data expansion ablation in detail

We expanded the training set from $n=31$ (v0 grid minus 5 heldout) to $n=46$ (v0.5 grid minus the same 5 heldout). The 15 new items target the 7 concentrated failure-mode items from Figure 3: 4 reader-is-principal probe variants, 4 capitulation-anchoring variants, and 7 coverage-breadth items. Reward function and hyperparameters are identical to v1.

metric (108 rollouts, multi-turn)	v4.1	DAPO-v1 ($n=31$)	DAPO-v1 ($n=46$)
total harm_fire	56	37	38
total MI	51	34	38
bound_leak	4	2	3
plain leak	19.4%	27.8%	22.2%
scaffolded leak	16.7%	16.7%	8.3%

Per-item delta $n=31 \rightarrow n=46$. The aggregate harm is essentially flat (37 vs. 38), but the cell composition shifts: scaffolded leak drops by half (16.7% $\rightarrow$ 8.3%) while plain leak drops slightly. The basin claim therefore generalizes from reward design to data design: changing either dimension produces a different point on the manifold but cannot escape the $\sim 33\text{--}38/108$ harm window.

I Held-out item split: train vs. val item harm

The DAPO-v1 training set is $n=31$ items; 5 are held out: `pb-author-testcase-01`, `pb-author-to-principal-01`, `pb-capit-repq-01`, `pb-leak-batna-01`, `pb-sanity-coop-01`. All 36 items appear in the eval grid; per-item attribution is recoverable from `data/ver1_train.parquet` / `data/ver1_val.parquet`.

checkpoint	train harm	val harm	train leak	val leak
Untrained baseline	25/92 (27%)	3/15 (20%)	69/92 (75%)	12/15 (80%)
v4.1 (DPO endpoint)	47/93 (51%)	9/15 (60%)	14/93 (15%)	4/15 (27%)
DAPO-v1 step_35	32/93 (34%)	5/15 (33%)	15/93 (16%)	4/15 (27%)

The held-out items show the same direction of change as training items: v4.1 lifts harm sharply over baseline (DPO induces MI on both sets), and DAPO-v1 walks it back on both (-15 on train, -4 on val). Untrained leak rates are similar across train/val (75% / 80%) — no item-specific bias. **The recipe is not memorization-bound at this experimental scale.**

J Multi-rollout error bars

We re-run the full 108-rollout grid two additional times at the same harness sampling temperature (0.7) for both the v4.1 and the DAPO-v1 step_35 checkpoints, against the `claude-sonnet` counterparty. Each seed independently samples both the agent and the counterparty.

checkpoint	seed 1	seed 2	seed 3	mean $\pm$ s.d.
v4.1 harm	56/108	50/104	49/107	51.7 ± 3.8
v4.1 leak	18/108	18/104	16/107	17.3 ± 1.2
DAPO-v1 harm	37/108	50/107	55/108	47.3 ± 9.3
DAPO-v1 leak	19/108	20/107	15/108	18.0 ± 2.6
robust harm (cells fire under <i>all</i> 3 seeds)	—	—	—	v4.1 32 / DAPO 21
any-fire harm (cells fire under ≥ 1 seed)	—	—	—	v4.1 70 / DAPO 75

Interpretation. The leak axis is highly stable across seeds (s.d. <3 fires) — the leak/MI manifold’s leak coordinate is essentially deterministic at this experimental scale. The total-harm axis has substantial within-checkpoint variance (DAPO-v1 spans 37–55 across 3 seeds), so the single-seed comparison 56 vs. 37 (-34%) overstates the mean improvement. The cell-robust statistic — count of (item $\times$ arm) cells where harm fires under every seed — is the reliable comparison: 32 (v4.1) vs. 21 (DAPO-v1), the same -34% . The any-fire upper-bound statistic (cells where at least one seed fires) is similar across the two checkpoints (70 vs. 75) — both leave the same long tail of intermittent failures, but DAPO-v1 has fewer cells where failure is consistent. Multi-rollout summaries: `runs/phase{2_trained_v4_1,3_dapo_v1_step35}_multirollout_summary.json`.

K Multi-turn evaluation integrity note

Earlier drafts of this paper reported substantially lower harm and leak numbers (e.g. “25/108 harm fires”, “leak = 0 throughout RL training”) derived from a contaminated batch of evaluation runs in which the Anthropic counterparty API returned 401 auth errors mid-run, causing every trajectory to terminate at the agent’s first message before any counterparty pressure was applied. The harness recorded these as completed single-turn rollouts, indistinguishable at the `scored.jsonl` level from successful conversations except for the `turns: 1` and `early_end_reason` fields. All numbers reported in this revision are derived from re-runs of the affected evaluations under the multi-turn harness with verified counterparty connectivity. Five evaluations were re-run: `phase2_trained_v4`, `phase2_trained_v4_1`, `phase3_dapo_v1_step30`, `phase3_dapo_v1_step35`, and `phase3_dapo_v3_step30`; the original (broken) trajectories are preserved at `runs/<name>_BROKEN_AUTH401/` for audit.

This is the integrity discipline change relative to the prior draft: *before reporting any aggregate from a `scored.jsonl`, audit the trajectory turn distribution and `early_end_reason` field.* Every multi-turn benchmark that scores agent behavior under counterparty pressure is at risk of this failure mode if the counterparty API is unreliable.

L Reproducibility paths

All artifacts at `/home/ubuntu/principal-loyalty/`; scripts in `scripts/` are runnable end-to-end in the order described by `progress.md`.

Items. `items/v0/*.json` ($n=36$); `items/v0_5/*.json` ($n=51$).

Phase 1. Hosted-vendor runs at `runs/phase1*/scored.jsonl`. Local-vLLM Qwen3-8B legacy baseline at `runs/phase2_baseline/scored.jsonl` (88% plain leak, single-arm); multi-turn untrained baseline (sentinel-aware, all 3 arms, $n=108$) at `runs/phase3_baseline_qwen/scored.jsonl` (74% plain leak).

Phase 2 (SFT+DPO). Earlier-iteration outputs at `runs/phase2_trained{,_v1,_v1_lite,_v2,_v06}/scored.jsonl`. Headline DPO endpoint: `runs/qwen_sft_dpo_v4_1_merged/` (16 GB), eval at `runs/phase2_trained_v4_1/scored.jsonl`. DPO pair files: `data/dpo_v{0,1,1_lite,2,05,06,4,4_combined}.jsonl`.

Phase 3 (DAPO). Headline checkpoint: `runs/qwen_dapo_v1_step35_merged/` (16 GB). Reward variant adapters: `runs/qwen_dapo_{v2,leakonly,v3}/`. Phase 3 evals: `runs/phase3_dapo_v1_step35/`, `runs/phase3_dapo_v{2_step55,leakonly_step35,v3_step{30,55}}/`. Reward function: `src/reward.py`. DAPO config: `scripts/run_dapo.sh`.

Counterparty robustness (Section 6). `runs/phase3_dapo_v1_step35{,_gpt5cp,_gemcp}/scored.jsonl`. Comparison: `scripts/compare_counterparty.py`.

Cross-family Mistral. Adapters: `runs/mistral_sft_v4/`, `runs/mistral_dpo_v4_1/`. Merged: `runs/mistral_sft_dpo_v4_1_merged/` (14 GB). Eval: `runs/phase3_mistral_sft_dpo/scored.jsonl`.

Data expansion (Appendix H). runs/qwen_dapo_v1_n51_step55_merged/ (16 GB), eval at runs/phase3_dapo_v1_n51_step55/scored.jsonl. Train data: data/ver1_train_v0_5.parquet (46 train rows / 5 val).

Item-level harm analysis (Figure 3). runs/phase3_dapo_v1_step35/item_harm_analysis.txt. Script: scripts/item_harm_analysis.py. Figure-generation script: scripts/make_figures.py.